

Variational Bayes for LC–MS Feature Detection

A mean-field model for elution profiles, isotope patterns and charge states

Summary. An LC–MS map is treated as a weighted point cloud in (retention time, m/z). Each pixel is generated either by one of K features or by a uniform background. A feature has a Gaussian elution profile in RT and an isotope comb in m/z, whose peak heights follow a binomial in the carbon count n_k and the ^{13}C probability p . RT and m/z are conditionally independent given the feature. We derive coordinate-ascent variational inference (CAVI) updates for all factors of a structured mean-field approximation.

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1 Model

1.1 Data

The map is a grid of non-empty pixels $n = 1, \dots, N$. Each pixel has a retention time t_n , an m/z value y_n and an intensity, converted to an **effective ion count** $w_n = \gamma I_n$. All w_n ions of a pixel share the same (t_n, y_n) , so every sum over ions becomes a weighted sum over pixels. The constant $\Delta = 1.00336$ Da is the mass difference between ^{13}C and ^{12}C .

1.2 Generative story

Every ion is produced as follows:

1. **Which feature?** $z_n \sim \text{Cat}(\pi)$, where $z_n = 0$ means background.

2. **Background** ($z_n = 0$): (t_n, y_n) is uniform over the window, $p(t_n, y_n \mid z_n = 0) = 1/(TY)$.
3. **Feature** $k \geq 1$:
 - **How many ^{13}C atoms?** $j_n \mid z_n = k \sim \mathcal{B}(n_k, p)$
 - **Retention time:** $t_n \mid z_n = k \sim \mathcal{N}(\mu_k, \sigma_k^2)$
 - **m/z:** $y_n \mid z_n = k, j_n \sim \mathcal{N}(m_k + j_n \Delta / c_k, s^2)$

The two main assumptions are that the isotope pattern is independent of RT **given the feature** (all isotopologues co-elute), and that the m/z peaks of all features share one global width s^2 .

UPDATE: RESPONSIBILITIES

We can setup the generative model:

$$p(t_i, y_i \mid z_i, j_i, \theta) = \begin{cases} \frac{\pi_0}{TY} & k = 0 \text{ (background)} \\ \mathcal{N}(t_i; \mu_k, \sigma_k^2) \mathcal{N}(y_i; m_k + \frac{j_i \Delta}{c_k}, s^2) & k \geq 1 \end{cases}$$

1.3 Graphical model

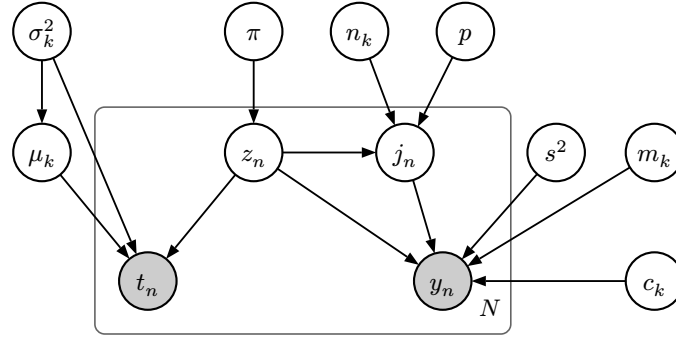


Figure 1: Graphical model. Shaded nodes are observed. Variables with index k exist once per feature (plate over K omitted); p and s^2 are global. Hyperparameters are not shown.

1.4 Variables

Symbol	Meaning	Scope	Type
z_n	feature assignment of pixel n ($z_n = 0$: background)	pixel	latent, categorical
j_n	isotope index: number of ^{13}C atoms of the ion	pixel	latent, discrete
t_n	retention time	pixel	observed
y_n	m/z	pixel	observed
w_n	effective ion count (weight)	pixel	observed
π_k	relative abundance of feature k (π_0 : background)	global	latent
μ_k	apex retention time	feature	latent
σ_k^2	RT variance (elution width)	feature	latent
m_k	monoisotopic m/z	feature	latent
c_k	charge, $c_k \in \{1, 2, 3, 4\}$	feature	latent, discrete
n_k	number of carbon atoms	feature	latent, discrete
p	^{13}C probability	global	latent
s^2	m/z peak variance	global	latent
Δ	$^{13}\text{C} - ^{12}\text{C}$ mass difference	—	constant

Table 1: Variables of the model.

2 Priors

All priors are (conditionally) conjugate, except for the carbon count, which lives on a finite grid.

Variable	Prior	Reason
π	$\text{Dir}(\alpha)$	conjugate to the categorical z ; $\alpha_k < 1$ prunes unused components
(μ_k, σ_k^2)	$\text{NIG}(m_{0k}, \nu_{0k}, a_0, b_0)$	conjugate for a Gaussian with unknown mean and variance
m_k	$\mathcal{N}(\widehat{m}_k, \tau_k^2)$	conjugate given c_k, j and s^2
c_k	$\text{Cat}(\rho)$ on $\{1, 2, 3, 4\}$	small discrete support, enumerated exactly
n_k	$p(n_k \mid \widehat{m}_k)$	no conjugate prior for the binomial's n ; exact on a grid
p	$\text{Beta}(a_p, b_p)$	conjugate to the binomial in p
s^2	$\text{IG}(a_s, b_s)$	conjugate for the variance of a Gaussian with known mean

Table 2: Priors at a glance.

The properties below are stated for generic parameters. They are used with the **variational** parameters (marked with a tilde) in the updates.

Mixing weights: $\pi \sim \text{Dir}(\alpha)$

The Dirichlet runs over $K + 1$ components including the background. Choosing $\alpha_k < 1$ lets VB push unused components towards zero weight, so one can start with a few more candidates than necessary. A larger α_0 gives the background more prior weight if a lot of noise is expected.

$$\log p(\pi) = \sum_{k=0}^K (\alpha_k - 1) \log \pi_k + \text{const}$$

$$\mathbb{E}[\log \pi_k] = \psi(\alpha_k) - \psi\left(\sum_{k'=0}^K \alpha_{k'}\right), \quad \mathbb{E}[\pi_k] = \frac{\alpha_k}{\sum_{k'=0}^K \alpha_{k'}}$$

RT profile: $(\mu_k, \sigma_k^2) \sim \text{NIG}(m_{0k}, \nu_{0k}, a_0, b_0)$

This means $\mu_k \mid \sigma_k^2 \sim \mathcal{N}(m_{0k}, \sigma_k^2 / \nu_{0k})$ and $\sigma_k^2 \sim \text{IG}(a_0, b_0)$. m_{0k} is the candidate's RT, ν_{0k} says how much we trust it, and a_0, b_0 encode the typical chromatographic peak width.

$$\begin{aligned} \log p(\mu_k, \sigma_k^2) &= -\frac{1}{2} \log \sigma_k^2 - \frac{\nu_{0k}(\mu_k - m_{0k})^2}{2\sigma_k^2} - (a_0 + 1) \log \sigma_k^2 - \frac{b_0}{\sigma_k^2} + \text{const} \\ &= -\left(a_0 + \frac{3}{2}\right) \log \sigma_k^2 - \frac{1}{\sigma_k^2} \left(\frac{\nu_{0k}(\mu_k - m_{0k})^2}{2} + b_0\right) + \text{const} \end{aligned}$$

For $\text{NIG}(m, \nu, a, b)$:

$$\mathbb{E}\left[\frac{1}{\sigma^2}\right] = \frac{a}{b}, \quad \mathbb{E}[\log \sigma^2] = \log b - \psi(a), \quad \mathbb{E}[\mu \mid \sigma^2] = m, \quad \text{Var}[\mu \mid \sigma^2] = \frac{\sigma^2}{\nu}$$

Monoisotopic m/z: $m_k \sim \mathcal{N}(\widehat{m}_k, \tau_k^2)$

Given c_k, j and s^2 , the m/z likelihood is Gaussian in m_k , so this prior is conjugate. $\widehat{m}_k$ is the candidate from the peak picker, and τ_k corresponds to a few ppm of $\widehat{m}_k$.

$$\log p(m_k) = -\frac{(m_k - \widehat{m}_k)^2}{2\tau_k^2} + \text{const}$$

Charge: $c_k \sim \text{Cat}(\rho)$ on $\{1, 2, 3, 4\}$

The support is small and discrete, so it is enumerated exactly. ρ reflects typical charge distributions; for tryptic peptides, 2+ dominates. The prior is shared, the posterior $q(c_k)$ is individual per feature.

$$\log p(c_k = c) = \log \rho_c, \quad \mathbb{E}[f(c_k)] = \sum_c q(c_k = c) f(c)$$

Carbon count: $n_k \sim p(n_k \mid \widehat{m}_k)$

An empirical categorical from an in-silico digest, evaluated at the candidate mass $\widehat{m}_k$. There is no conjugate prior for the number of trials of a binomial, but the grid is exact and cheap. Conditioning on the candidate mass (not on the latent m_k) keeps $q(m_k)$ Gaussian. Expectations are finite sums:

$$\mathbb{E}[n_k] = \sum_C q(n_k = C) C, \quad \mathbb{E}\left[\log\left(\binom{n_k}{j}\right)\right] = \sum_C q(n_k = C) \log\left(\binom{C}{j}\right)$$

Heavy-isotope probability: $p \sim \text{Beta}(a_p, b_p)$

The mean $a_p/(a_p + b_p) \approx 0.0107$ is the natural ^{13}C abundance, and a large $a_p + b_p$ makes the prior tight. This also resolves the n - p trade-off.

$$\begin{aligned} \log p(p) &= (a_p - 1) \log p + (b_p - 1) \log(1 - p) + \text{const} \\ \mathbb{E}[\log p] &= \psi(a_p) - \psi(a_p + b_p), \quad \mathbb{E}[\log(1 - p)] = \psi(b_p) - \psi(a_p + b_p) \end{aligned}$$

m/z peak width: $s^2 \sim \text{IG}(a_s, b_s)$

The prior mean comes from the instrument's resolution or mass accuracy.

$$\begin{aligned} \log p(s^2) &= -(a_s + 1) \log s^2 - \frac{b_s}{s^2} + \text{const} \\ \mathbb{E}\left[\frac{1}{s^2}\right] &= \frac{a_s}{b_s}, \quad \mathbb{E}[\log s^2] = \log b_s - \psi(a_s) \end{aligned}$$

3 Variational approximation

3.1 Mean-field family

We approximate the posterior with the structured mean-field family

$$q = q(z, j) q(\pi) q(\mu, \sigma^2) q(m, c) q(n) q(p) q(s^2).$$

z and j are kept jointly, because the isotope index only has a meaning relative to a feature. m and c are kept jointly, because the charge sets the comb spacing and hence the best mass. The factorizations over pixels n and features k are not assumptions: they follow from the model (induced factorizations).

3.2 Coordinate ascent

Since

$$q^* = \arg \min_{q \in \mathcal{Q}} D_{\text{KL}}(q(\theta) \parallel p(\theta \mid x)) = \arg \max_{q \in \mathcal{Q}} \mathcal{L}(q),$$

the optimal factor q_i , with all other factors fixed, is

$$\log q_i^*(\theta_i) = \mathbb{E}_{q_{-i}}[\log p(x, \theta)] + \text{const},$$

where the expectation runs over all factors except q_i . Each update maximizes the ELBO with respect to one factor, so the ELBO increases monotonically.

Because:

$$\begin{aligned}
\mathcal{L}(q_i) &= \mathbb{E}_{q,j} \left[\log \frac{p(x, z_j)}{q(z_j)} \right] \\
&= \mathbb{E}_{q,j} [\log p(x, z_j)] + \sum_i \mathbb{H}[q_i] \\
&= \mathbb{E}_{q_i} [\mathbb{E}_{q,j \neq i} [\log p(x, z_j)]] + \mathbb{H}[q_i] + \sum_{i \neq j} \mathbb{H}[q_j] \\
&= \mathbb{E}_{q_i} [\mathbb{E}_{q,j \neq i} [\log p(x, z_j)]] + \mathbb{H}[q_i] + \sum_{i \neq j} \mathbb{H}[q_j] \\
&= \mathbb{E}_{q_i} [\tilde{p}(x, z_i)] + \mathbb{H}[q_i] + \sum_{i \neq j} \mathbb{H}[q_j] \\
&= D_{\text{KL}}(q_i \parallel \tilde{p}(x, z_i)) + \text{const}
\end{aligned}$$

3.3 Log joint

$$\begin{aligned}
\log p &= \underbrace{\log p(z \mid \pi) + \log p(j \mid z, n, p) + \log p(t \mid z, \mu, \sigma^2) + \log p(y \mid z, j, m, s^2, c)}_{\text{likelihood terms}} \\
&\quad + \underbrace{\log p(\pi) + \log p(\mu, \sigma^2) + \log p(m) + \log p(c) + \log p(n) + \log p(p) + \log p(s^2)}_{\text{priors}}
\end{aligned}$$

Each update only needs the terms that contain its variable (its Markov blanket); everything else goes into the constant.

3.4 Notation and identities

Symbol	Definition	Meaning
r_{nkj}	$\mathbb{E}[z_{nk} \mathbb{1}[j_n = j]]$	responsibility of sub-peak (k, j) for pixel n
r_{n0}	$\mathbb{E}[z_{n0}]$	responsibility of the background
R_{nk}	$\sum_j r_{nkj}$	responsibility of feature k
N_k	$\sum_n w_n R_{nk}$	expected number of ions of feature k
λ	$\mathbb{E}[1/s^2] = \tilde{a}_s/\tilde{b}_s$	expected m/z precision
$u_{nj}(c)$	$y_n - j\Delta/c$	monoisotopic m/z implied by pixel n as isotope j at charge c
S	$\sum_n \sum_k \sum_j w_n r_{nkj} j$	expected total number of ^{13}C atoms

Table 3: Shorthand used in the updates.

The following identities are used repeatedly:

Expected square $\mathbb{E}[(X - a)^2] = \text{Var}[X] + (\mathbb{E}[X] - a)^2$ for any random variable X and constant a .

Tower rule $\mathbb{E}_{q(x,v)}[f] = \mathbb{E}_{q(v)}[\mathbb{E}_{q(x|v)}[f]]$.

Assignments $\mathbb{E}_{q(z,j)}[z_{nk} f(j_n)] = \sum_j r_{nkj} f(j)$.

Completing the square with weights ω_i and centers u_i , and $A = \sum_i \omega_i$, $B = \sum_i \omega_i u_i$, $C = \sum_i \omega_i u_i^2$:

$$\sum_i \omega_i (x - u_i)^2 = A \left(x - \frac{B}{A} \right)^2 + \left(C - \frac{B^2}{A} \right), \quad C - \frac{B^2}{A} = \sum_i \omega_i \left(u_i - \frac{B}{A} \right)^2.$$

Gaussian integral $\int \exp\left(-\frac{A}{2}(x - x_0)^2\right) dx = \sqrt{2\pi/A}$.

Numerics. $C - B^2/A$ is the difference of two huge, nearly equal numbers (squared m/z values around 10^5 times a large precision). Always evaluate it in the deviation form $\sum_i \omega_i (u_i - B/A)^2$, which only sums small positive terms. Normalize all discrete distributions with log-sum-exp.

4 Updates

4.1 Assignments: $q(z, j)$

Collecting the four likelihood terms and taking expectations over all other factors:

$$\begin{aligned} \log q^*(z, j) &= \mathbb{E}_{q(\pi)}[\log p(z \mid \pi)] + \mathbb{E}_{q(n), q(p)}[\log p(j \mid z, n, p)] \\ &\quad + \mathbb{E}_{q(\mu, \sigma^2)}[\log p(t \mid z, \mu, \sigma^2)] + \mathbb{E}_{q(m, c), q(s^2)}[\log p(y \mid z, j, m, s^2, c)] + \text{const} \\ &= \sum_n \left(z_{n0} \log \rho_{n0} + \sum_{k=1}^K \sum_j z_{nk} \mathbb{1}[j_n = j] \log \rho_{nkj} \right) + \text{const}, \end{aligned}$$

with one term per sub-peak (k, j) :

$$\begin{aligned} \log \rho_{nkj} &= \mathbb{E}[\log \pi_k] + \mathbb{E}[\log \mathcal{B}(j \mid n_k, p)] \\ &\quad + \mathbb{E}[\log \mathcal{N}(t_n \mid \mu_k, \sigma_k^2)] + \mathbb{E}[\log \mathcal{N}(y_n \mid m_k + j\Delta/c_k, s^2)]. \end{aligned}$$

We evaluate the four expectations one by one.

Mixing weight. From the Dirichlet:

$$\mathbb{E}[\log \pi_k] = \psi(\tilde{\alpha}_k) - \psi\left(\sum_{k'=0}^K \tilde{\alpha}_{k'}\right).$$

Isotope pattern. Expanding the binomial, and using that n_k and p are independent under q :

$$\begin{aligned} \mathbb{E}[\log \mathcal{B}(j \mid n_k, p)] &= \mathbb{E}\left[\log \binom{n_k}{j}\right] + j \mathbb{E}[\log p] + (\mathbb{E}[n_k] - j) \mathbb{E}[\log(1 - p)] \\ &= \sum_C q(n_k = C) \log \binom{C}{j} + j(\psi(\tilde{\alpha}_p) - \psi(\tilde{b}_p)) + \mathbb{E}[n_k](\psi(\tilde{b}_p) - \psi(\tilde{\alpha}_p + \tilde{b}_p)). \end{aligned}$$

Retention time. μ_k and σ_k^2 are dependent under q , so we average over $\mu_k \mid \sigma_k^2$ first:

$$\begin{aligned} \mathbb{E}[\log \mathcal{N}(t_n \mid \mu_k, \sigma_k^2)] &= -\frac{1}{2} \left(\log 2\pi + \mathbb{E}[\log \sigma_k^2] + \mathbb{E}_{q(\sigma_k^2)} \left[\frac{1}{\sigma_k^2} \mathbb{E}_{q(\mu_k \mid \sigma_k^2)} [(t_n - \mu_k)^2] \right] \right) \\ &= -\frac{1}{2} \left(\log 2\pi + \mathbb{E}[\log \sigma_k^2] + \mathbb{E}_{q(\sigma_k^2)} \left[\frac{1}{\sigma_k^2} \left(\frac{\sigma_k^2}{\tilde{\nu}_k} + (t_n - \tilde{\mu}_k)^2 \right) \right] \right) \\ &= -\frac{1}{2} \left(\log 2\pi + \log \tilde{b}_k - \psi(\tilde{\alpha}_k) + \frac{1}{\tilde{\nu}_k} + \frac{\tilde{\alpha}_k}{\tilde{b}_k} (t_n - \tilde{\mu}_k)^2 \right). \end{aligned}$$

m/z. s^2 is independent of (m_k, c_k) . For the squared residual we use the tower rule over the charge and the expected square; the conditional variance $\tilde{\tau}_k^2$ does not depend on c (see Section 4.4):

$$\begin{aligned} \mathbb{E}[\log \mathcal{N}(y_n \mid m_k + j\Delta/c_k, s^2)] &= -\frac{1}{2} \left(\log 2\pi + \mathbb{E}[\log s^2] + \mathbb{E}\left[\frac{1}{s^2}\right] \mathbb{E}_{q(m_k, c_k)} \left[\left(m_k + \frac{j\Delta}{c_k} - y_n \right)^2 \right] \right) \\ &= -\frac{1}{2} \left(\log 2\pi + \log \tilde{b}_s - \psi(\tilde{\alpha}_s) + \lambda \left(\tilde{\tau}_k^2 + \sum_c q(c_k = c) \left(\tilde{m}_{kc} + \frac{j\Delta}{c} - y_n \right)^2 \right) \right). \end{aligned}$$

Background. No RT, m/z or isotope structure:

$$\log \rho_{n0} = \psi(\tilde{\alpha}_0) - \psi\left(\sum_{k'=0}^K \tilde{\alpha}_{k'}\right) - \log(TY).$$

Constants. The $\log 2\pi$ terms are the same for every feature, but they differ from the background's $\log(TY)$. They must therefore be kept when the background is part of the normalization.

Exponentiating gives a categorical distribution over the sub-peaks of each pixel plus the background:

$$q^*(z, j) \propto \prod_n \left(\rho_{n0}^{z_{n0}} \prod_{k=1}^K \prod_j \rho_{nkj}^{z_{nk} \mathbb{1}[j_n=j]} \right).$$

UPDATE: RESPONSIBILITIES

$$r_{nkj} = \frac{\rho_{nkj}}{\rho_{n0} + \sum_{k'=1}^K \sum_{j'} \rho_{nk'j'}}, \quad r_{n0} = \frac{\rho_{n0}}{\rho_{n0} + \sum_{k'=1}^K \sum_{j'} \rho_{nk'j'}}, \quad R_{nk} = \sum_j r_{nkj}$$

Computed in log space with log-sum-exp. The pixel weights w_n do not enter here.

4.2 Mixing weights: $q(\pi)$

Only $\log p(z \mid \pi)$ and the prior contain π . With $R_{n0} := r_{n0}$:

$$\begin{aligned} \log q^*(\pi) &= \mathbb{E}_{q(z)}[\log p(z \mid \pi)] + \log p(\pi) + \text{const} \\ &= \sum_n \sum_{k=0}^K w_n R_{nk} \log \pi_k + \sum_{k=0}^K (\alpha_k - 1) \log \pi_k + \text{const} \\ &= \sum_{k=0}^K (\alpha_k + N_k - 1) \log \pi_k + \text{const}. \end{aligned}$$

Matching with $\log \text{Dir}(\pi \mid \tilde{\alpha}) = \sum_k (\tilde{\alpha}_k - 1) \log \pi_k + \text{const}$:

UPDATE: DIRICHLET

$$q^*(\pi) = \text{Dir}(\tilde{\alpha}), \quad \tilde{\alpha}_k = \alpha_k + N_k, \quad k = 0, \dots, K$$

4.3 RT profiles: $q(\mu, \sigma^2)$

The expression is a sum over features, so $q^*(\mu, \sigma^2) = \prod_k q^*(\mu_k, \sigma_k^2)$. For one feature:

$$\begin{aligned} \log q^*(\mu_k, \sigma_k^2) &= \sum_n w_n R_{nk} \left(-\frac{1}{2} \log \sigma_k^2 - \frac{1}{2\sigma_k^2} (t_n - \mu_k)^2 \right) + \log p(\mu_k, \sigma_k^2) + \text{const} \\ &= -\left(\frac{N_k}{2} + \frac{1}{2} + a_0 + 1 \right) \log \sigma_k^2 \\ &\quad - \frac{1}{\sigma_k^2} \left(b_0 + \frac{1}{2} \left[\nu_{0k} (\mu_k - m_{0k})^2 + \sum_n w_n R_{nk} (t_n - \mu_k)^2 \right] \right) + \text{const}. \end{aligned}$$

Completing the square. The bracket is a weighted sum of squares with centers m_{0k} (weight ν_{0k}) and t_n (weights $w_n R_{nk}$):

$$\begin{aligned}
& \nu_{0k}(\mu_k - m_{0k})^2 + \sum_n w_n R_{nk}(t_n - \mu_k)^2 \\
&= \underbrace{(\nu_{0k} + N_k)\mu_k^2}_A - 2 \underbrace{\left(\nu_{0k}m_{0k} + \sum_n w_n R_{nk}t_n\right)}_B \mu_k + \underbrace{\left(\nu_{0k}m_{0k}^2 + \sum_n w_n R_{nk}t_n^2\right)}_C \\
&= A\left(\mu_k - \frac{B}{A}\right)^2 + \left(C - \frac{B^2}{A}\right).
\end{aligned}$$

Splitting into Gaussian and inverse-gamma. The Gaussian over $\mu_k \mid \sigma_k^2$ needs its own $-\frac{1}{2} \log \sigma_k^2$ for normalization; the rest belongs to the inverse-gamma:

$$\log q^*(\mu_k, \sigma_k^2) = \underbrace{-\frac{1}{2} \log \sigma_k^2 - \frac{A}{2\sigma_k^2} \left(\mu_k - \frac{B}{A}\right)^2}_{\log \mathcal{N}(\mu_k \mid B/A, \sigma_k^2/A)} - \underbrace{\left(a_0 + \frac{N_k}{2} + 1\right) \log \sigma_k^2 - \frac{1}{\sigma_k^2} \left(b_0 + \frac{1}{2} \left(C - \frac{B^2}{A}\right)\right)}_{\log \text{IG}(\sigma_k^2)} + \text{const.}$$

UPDATE: NORMAL-INVERSE-GAMMA

$$\begin{aligned}
q^*(\mu_k, \sigma_k^2) &= \text{NIG}(\tilde{\mu}_k, \tilde{\nu}_k, \tilde{a}_k, \tilde{b}_k) \\
\tilde{\nu}_k &= \nu_{0k} + N_k, \quad \tilde{\mu}_k = \frac{\nu_{0k}m_{0k} + \sum_n w_n R_{nk}t_n}{\nu_{0k} + N_k}, \\
\tilde{a}_k &= a_0 + \frac{N_k}{2}, \quad \tilde{b}_k = b_0 + \frac{1}{2} \left(\nu_{0k}(m_{0k} - \tilde{\mu}_k)^2 + \sum_n w_n R_{nk}(t_n - \tilde{\mu}_k)^2 \right).
\end{aligned}$$

The last form of $\tilde{b}_k$ equals $b_0 + \frac{1}{2}(C - B^2/A)$ and is numerically stable.

4.4 Mass and charge: $q(m, c)$

The terms with $\log s^2$ do not depend on (m, c) and go into the constant. With $\lambda = \mathbb{E}[1/s^2]$:

$$\begin{aligned}
\log q^*(m, c) &= \mathbb{E}_{q(z, j), q(s^2)} [\log p(y \mid z, j, m, s^2, c)] + \log p(m) + \log p(c) + \text{const} \\
&= - \sum_n \sum_k \mathbb{E} \left[\frac{1}{2s^2} \right] \mathbb{E}_{q(z, j)} \left[z_{nk} \left(m_k + \frac{j_n \Delta}{c_k} - y_n \right)^2 \right] + \log p(m) + \log p(c) + \text{const} \\
&= \sum_k \left(-\frac{\lambda}{2} \sum_n \sum_j w_n r_{nkj} \left(m_k + \frac{j_n \Delta}{c_k} - y_n \right)^2 - \frac{(m_k - \hat{m}_k)^2}{2\tau_k^2} + \log \rho_{c_k} \right) + \text{const.}
\end{aligned}$$

It factorizes over features. For one feature and one **fixed charge hypothesis** $c_k = c$, rewrite the residual with the implied monoisotopic position $u_{nj}(c) = y_n - j\Delta/c$:

$$\begin{aligned}
& \lambda \sum_n \sum_j w_n r_{nkj} (m_k - u_{nj}(c))^2 + \frac{1}{\tau_k^2} (m_k - \hat{m}_k)^2 \\
&= \underbrace{\left(\lambda N_k + \frac{1}{\tau_k^2} \right) m_k^2}_A - 2 \underbrace{\left(\lambda \sum_n \sum_j w_n r_{nkj} u_{nj}(c) + \frac{\hat{m}_k}{\tau_k^2} \right) m_k}_{B_c} + \underbrace{\left(\lambda \sum_n \sum_j w_n r_{nkj} u_{nj}(c)^2 + \frac{\hat{m}_k^2}{\tau_k^2} \right)}_{C_c} \\
&= A \left(m_k - \frac{B_c}{A} \right)^2 + \left(C_c - \frac{B_c^2}{A} \right).
\end{aligned}$$

The weights do not depend on c , only the centers do; hence A is the same for every charge. Putting the $-\frac{1}{2}$ and the charge prior back:

$$\log q^*(m_k, c_k = c) = -\frac{A}{2} \left(m_k - \frac{B_c}{A} \right)^2 - \frac{1}{2} \left(C_c - \frac{B_c^2}{A} \right) + \log \rho_c + \text{const.}$$

Conditional on the charge, this is a Gaussian in m_k with precision A and mean B_c/A .

Marginal over the mass. Integrating m_k out with the Gaussian integral:

$$\begin{aligned} q^*(c_k = c) &= \int q^*(m_k, c_k = c) dm_k \propto \rho_c \exp \left(-\frac{1}{2} \left(C_c - \frac{B_c^2}{A} \right) \right) \sqrt{\frac{2\pi}{A}} \\ \Rightarrow \log q^*(c_k = c) &= \log \rho_c - \frac{1}{2} \left(C_c - \frac{B_c^2}{A} \right) + \text{const}, \end{aligned}$$

since A does not depend on c .

UPDATE: MASS AND CHARGE

$$q^*(m_k, c_k) = q^*(c_k) \mathcal{N}(m_k \mid \tilde{m}_{kc}, \tilde{\tau}_k^2)$$

$$\tilde{\tau}_k^2 = \frac{1}{\lambda N_k + 1/\tau_k^2}, \quad \tilde{m}_{kc} = \tilde{\tau}_k^2 \left(\lambda \sum_n \sum_j w_n r_{nkj} \left(y_n - \frac{j\Delta}{c} \right) + \frac{\hat{m}_k}{\tau_k^2} \right)$$

$$q^*(c_k = c) = \frac{\exp(\ell_c)}{\sum_{c'} \exp(\ell_{c'})}, \quad \ell_c = \log \rho_c - \frac{1}{2} \left(\lambda \sum_n \sum_j w_n r_{nkj} \left(y_n - \frac{j\Delta}{c} - \tilde{m}_{kc} \right)^2 + \frac{(\hat{m}_k - \tilde{m}_{kc})^2}{\tau_k^2} \right)$$

ℓ_c is written in the stable deviation form of $C_c - B_c^2/A$. It measures how well the comb with spacing Δ/c lines up with the data.

4.5 m/z peak width: $q(s^2)$

s^2 is global, so the sums run over all features $k \geq 1$. The inner expectation is the same as in the $q(z, j)$ update:

$$\begin{aligned} \log q^*(s^2) &= \mathbb{E}_{q(z, j), q(m, c)} [\log p(y \mid z, j, m, s^2, c)] + \log p(s^2) + \text{const} \\ &= - \sum_n \sum_k w_n R_{nk} \frac{1}{2} \log s^2 - \frac{1}{2s^2} \sum_n \sum_k \sum_j w_n r_{nkj} \mathbb{E}_{q(m_k, c_k)} \left[\left(m_k + \frac{j\Delta}{c_k} - y_n \right)^2 \right] \\ &\quad - (a_s + 1) \log s^2 - \frac{b_s}{s^2} + \text{const} \\ &= - \left(a_s + 1 + \frac{1}{2} \sum_{k=1}^K N_k \right) \log s^2 \\ &\quad - \frac{1}{s^2} \left(b_s + \frac{1}{2} \sum_n \sum_k \sum_j w_n r_{nkj} \left(\tilde{\tau}_k^2 + \sum_c q(c_k = c) \left(\tilde{m}_{kc} + \frac{j\Delta}{c} - y_n \right)^2 \right) \right) + \text{const}. \end{aligned}$$

UPDATE: INVERSE-GAMMA

$$\tilde{a}_s = a_s + \frac{1}{2} \sum_{k=1}^K N_k,$$

$$\tilde{b}_s = b_s + \frac{1}{2} \sum_n \sum_k \sum_j w_n r_{nkj} \left(\tilde{\tau}_k^2 + \sum_c q(c_k = c) \left(\tilde{m}_{kc} + \frac{j\Delta}{c} - y_n \right)^2 \right), \quad \lambda = \frac{\tilde{a}_s}{\tilde{b}_s}.$$

4.6 Carbon counts: $q(n)$

The term $j \log p$ does not contain n_k and goes into the constant; so does $-j \log(1-p)$. The prior is a separate table per feature, $\log p(n) = \sum_k \log p(n_k | \hat{m}_k)$:

$$\begin{aligned} \log q^*(n) &= \mathbb{E}_{q(z,j),q(p)}[\log p(j | z, n, p)] + \log p(n) + \text{const} \\ &= \sum_n \sum_k \sum_j w_n r_{nkj} \left(\log \binom{n_k}{j} + j \mathbb{E}[\log p] + (n_k - j) \mathbb{E}[\log(1-p)] \right) + \log p(n) + \text{const} \\ &= \sum_k \left(\sum_n \sum_j w_n r_{nkj} \log \binom{n_k}{j} + n_k N_k \mathbb{E}[\log(1-p)] + \log p(n_k | \hat{m}_k) \right) + \text{const}. \end{aligned}$$

There is no parametric family to recognize: $q(n_k)$ is evaluated on the grid.

UPDATE: CARBON COUNT (GRID)

$$\begin{aligned} \ell_k(C) &= \log p(n_k = C | \hat{m}_k) + \sum_n \sum_j w_n r_{nkj} \log \binom{C}{j} + C N_k (\psi(\tilde{b}_p) - \psi(\tilde{a}_p + \tilde{b}_p)) \\ q^*(n_k = C) &= \frac{\exp(\ell_k(C))}{\sum_{C'} \exp(\ell_k(C'))} \end{aligned}$$

The grid starts at the largest allowed isotope index, since $\binom{C}{j} = 0$ for $C < j$.

4.7 Heavy-isotope probability: $q(p)$

The binomial coefficient does not contain p and goes into the constant. $\mathbb{E}[n_k]$ does not depend on n or j , and $\sum_n \sum_j w_n r_{nkj} = N_k$:

$$\begin{aligned} \log q^*(p) &= \mathbb{E}_{q(z,j),q(n)}[\log p(j | z, n, p)] + \log p(p) + \text{const} \\ &= \sum_n \sum_k \sum_j w_n r_{nkj} (j \log p + (\mathbb{E}[n_k] - j) \log(1-p)) \\ &\quad + (a_p - 1) \log p + (b_p - 1) \log(1-p) + \text{const} \\ &= (a_p + S - 1) \log p + \left(b_p + \sum_k N_k \mathbb{E}[n_k] - S - 1 \right) \log(1-p) + \text{const}. \end{aligned}$$

UPDATE: BETA

$$q^*(p) = \text{Beta}(\tilde{a}_p, \tilde{b}_p), \quad \tilde{a}_p = a_p + S, \quad \tilde{b}_p = b_p + \sum_k N_k \mathbb{E}[n_k] - S$$

$\tilde{a}_p$ counts the expected ^{13}C atoms, $\tilde{b}_p$ the expected ^{12}C atoms.

5 Algorithm

Factor	Family	Variational parameters
$q(z_n, j_n)$	categorical over (k, j) and background	r_{nkj}, r_{n0}
$q(\pi)$	Dirichlet	$\tilde{\alpha}_0, \dots, \tilde{\alpha}_K$
$q(\mu_k, \sigma_k^2)$	Normal-inverse-gamma	$\tilde{\mu}_k, \tilde{\nu}_k, \tilde{a}_k, \tilde{b}_k$
$q(m_k c_k)$	Gaussian per charge	$\tilde{m}_{kc}, \tilde{\tau}_k^2$
$q(c_k)$	categorical on $\{1, 2, 3, 4\}$	$q(c_k = c)$
$q(n_k)$	categorical on a grid	$q(n_k = C)$
$q(p)$	Beta	$\tilde{a}_p, \tilde{b}_p$
$q(s^2)$	inverse-gamma	$\tilde{a}_s, \tilde{b}_s$

Table 4: Variational factors and their parameters.

Algorithm 1: CAVI from random features

```

1: procedure CAVI(pixels, K, J, I)
2:    $w_n \leftarrow I_n / \min_m I_m$ 
3:   for  $k = 1, \dots, K$  do
4:      $(m_{0k}, \hat{m}_k) \leftarrow (t_n, y_n)$  of a pixel  $n \sim w$ , with  $y_n \in [300, 2000]$ 
5:      $c_k \leftarrow \sim (0.1, 0.5, 0.3, 0.1)$ 
6:   end
7:    $q \leftarrow p(\theta)$ 
8:
9:   for  $i = 1, \dots, I$  do
10:     $r_{nkj}, r_{n0} \leftarrow$  softmax over  $(k, j)$  and background of  $\log \rho$ 
11:     $N_k, N_0 \leftarrow \sum_n w_n \sum_j r_{nkj}, \sum_n w_n r_{n0}$ 
12:     $q(\pi) \leftarrow \text{Dir}(\alpha + (N_0, \dots, N_K))$ 
13:     $q(\mu_k, \sigma_k^2) \leftarrow$  NIG update
14:     $q(m_k | c_k = c) \leftarrow$  Gaussian update    $q(c_k) \leftarrow$  softmax of  $\ell_c$ 
15:     $q(s^2) \leftarrow$  inverse-gamma update
16:     $q(n_k) \leftarrow$  grid update    $q(p) \leftarrow$  Beta update
17:     $\mathcal{L}_i \leftarrow$  ELBO of  $(r, q)$ 
18:   end
19:   return  $q, (\mathcal{L}_i)_i$ 
20: end

```

Prior	Value
position $(m_{0k}, \hat{m}_k)$	pixel drawn with probability $\propto w_n$ (uniform over the window: --init uniform)
charge	$c_k \sim (0.1, 0.5, 0.3, 0.1)$ (centres the average $p(n_k)$); the prior over c is the same for all k
$\tau_k, \nu_{0k}, \mathbb{E}[\sigma_k^2]$	0.1 Da, 0.1, (0.5 min) ²
$\mathbb{E}[s^2], \alpha_k, K$	(0.05 Da) ² (learned), 1, 30

Table 5: Random start (random_hyperparameters).

Efficiency. Only pixels inside a feature’s bounding box ($\kappa \approx 5$ widths around the apex and each comb tooth) get non-zero responsibilities for that feature. Pixels outside all boxes are pure background and only contribute their total weight to $\tilde{\alpha}_0$. Recompute the boxes every few iterations, not within a sweep.

Remark. The core issue is that almost all ions will belong to the background so $z_n = 0$. We do not want to calculate CAVI updates for each of those update steps. So we do not want to consider each component k for each isotope but only a fixed number of components k_{eff} so

$$\sum_n \sum_k \sum_j \dots = \sum_n \sum_{k_{\text{eff}}} \sum_n \dots$$

We can achieve that, via selecting a number of potential candidates first fitting those and then fitting the rest of not easy to fit features.

6 ELBO

$$\begin{aligned} \mathcal{L} = & \sum_n w_n \left[\sum_{k,j} r_{nkj} (\log \rho_{nkj} - \log r_{nkj}) + r_{n0} (\log \rho_{n0} - \log r_{n0}) \right] \\ & - D_{\text{KL}}(q(\pi) \parallel p(\pi)) - D_{\text{KL}}(q(p) \parallel p(p)) - D_{\text{KL}}(q(s^2) \parallel p(s^2)) \\ & - \sum_k \left[D_{\text{KL}}(q(\mu_k, \sigma_k^2) \parallel p) + D_{\text{KL}}(q(m_k, c_k) \parallel p) + D_{\text{KL}}(q(n_k) \parallel p) \right] \end{aligned}$$

Factor	KL divergence to the prior (elbo.py)
π	Dirichlet: $\log \Gamma(\sum \tilde{\alpha}) - \sum \log \Gamma(\tilde{\alpha}_k) - \dots + \sum (\tilde{\alpha}_k - \alpha_k)(\psi(\tilde{\alpha}_k) - \psi(\sum \tilde{\alpha}))$
(μ_k, σ_k^2)	$D_{\text{KL}}(\text{IG} \parallel \text{IG}) + \frac{1}{2} \left[\log(\tilde{\nu}_k / \nu_{0k}) + \nu_{0k} / \tilde{\nu}_k - 1 + \nu_{0k} (\tilde{\mu}_k - m_{0k})^2 \tilde{a}_k / \tilde{b}_k \right]$
(m_k, c_k)	$D_{\text{KL}}(\text{Cat}(q(c_k)) \parallel \text{Cat}(\rho_k)) + \sum_c q(c_k = c) D_{\text{KL}}(\mathcal{N}(\tilde{m}_{kc}, \tilde{\tau}_k^2) \parallel \mathcal{N}(\hat{m}_k, \tau_k^2))$
n_k	categorical on the grid
p	Beta
s^2	inverse-gamma (the gamma KL with shape a , rate b)

Table 6: KL terms. $\log \rho$ is taken at the current q ; each update raises $\mathcal{L}$.

7 Plots and results

`make_algorithm` (`plots/plot_cavi.py`) runs the algorithm on a spectrum and writes the plots below to `plots/cavi_real`. Colours identify components in all panels (Figure 3, last page); only the 12 components with most ions at the end are drawn. A component counts as supported if $N_k = \tilde{\alpha}_k - \alpha_k > 30$.

Plot	Panel	Meaning
elbo.png	$\mathcal{L}$ per sweep	lower bound on the log evidence; flattening means converged
	$ \Delta\mathcal{L} $, log	size of each step; crosses mark decreases (<code>float32</code> noise)
metrics.png	1 expected ions N_k	weighted number of pixels assigned to a component; dashed: support threshold
	2 supported components	left: how many of the K the data use; right: share of all ions explained by features, not background
	3 RT apex μ_k	posterior mean elution apex and ± 2 sd; dotted: reference RT of a matched component
	4 mass minus final	convergence of $\tilde{m}_{kc}$ (zero at the end by construction, not an accuracy); band: $\pm 2\tilde{\tau}_k$
	5 peak width	$\sqrt{b_s/(a_s - 1)}$, one value for all; how close in m/z a pixel must be to a component
	6 ^{13}C probability p	with 90% interval; near 0.011 the isotope teeth are explained by the comb, near 0 by components of their own
	7 final $q(c_k)$	charge posterior per component; a row equal to the prior is uninformed; crosses: reference charge
	8 final $q(n_k)$	carbon-count posterior ($n \approx 0.044 \cdot m \cdot c$); width: uncertainty; informative only where the comb is used
intensity.png	left	fitted against reference intensity of each feature, log axes; dashed $y = x$, solid median ratio, dotted ceiling; colour: components per feature; title: Pearson (log), Spearman
	right	the same as a ratio per feature, components above the bars, “none”: no component
fit_full.mp4	top	actual map, fitted expected map (posterior mean), posterior std of the expected counts; one frame per sweep, frame 0 is the random start
	bottom	projections on m/z and RT: actual (grey) against fitted (colour: std)
fit_component<k>.mp4		the same, zoomed on the component with most ions
..._3d.mp4		surface: fitted mean, colour: std, wireframe: actual counts

Table 7: The plots of `plot_cavi.py`.

Fitted map. With $W = \sum_n w_n$, $P_{kj} = \sum_n q(n_k = n) \text{Binom}(j \mid n, p)$, $\mu_{kcj} = m_{kc} + j\Delta/c$, the expected ions in the bin $[t_a, t_b] \times [y_a, y_b]$ are

$$W \left(\pi_0 \frac{\Delta t \Delta y}{TY} + \sum_k \pi_k A_k \sum_j P_{kj} \sum_c q(c_k = c) B_{kcj} \right)$$

$$A_k = \Phi \left(\frac{t_b - \mu_k}{\sigma_k} \right) - \Phi \left(\frac{t_a - \mu_k}{\sigma_k} \right), \quad B_{kcj} = \Phi \left(\frac{y_b - \mu_{kcj}}{s} \right) - \Phi \left(\frac{y_a - \mu_{kcj}}{s} \right).$$

The std is taken over 12 parameter draws from q ($q(c)$, $q(n)$ marginalised): the uncertainty of the model, not the noise.

Intensity. The intensity of a reference feature is $\min I$ times the expected ions of all components whose mass lies on one of its isotope positions $m + j\Delta/c$ ($j < J$, reference charge) with RT within 1 min (cavi/evaluate.py). The ceiling is (file intensity) / (sum of reference intensities): the reference is a model estimate and can exceed the pixel sum.

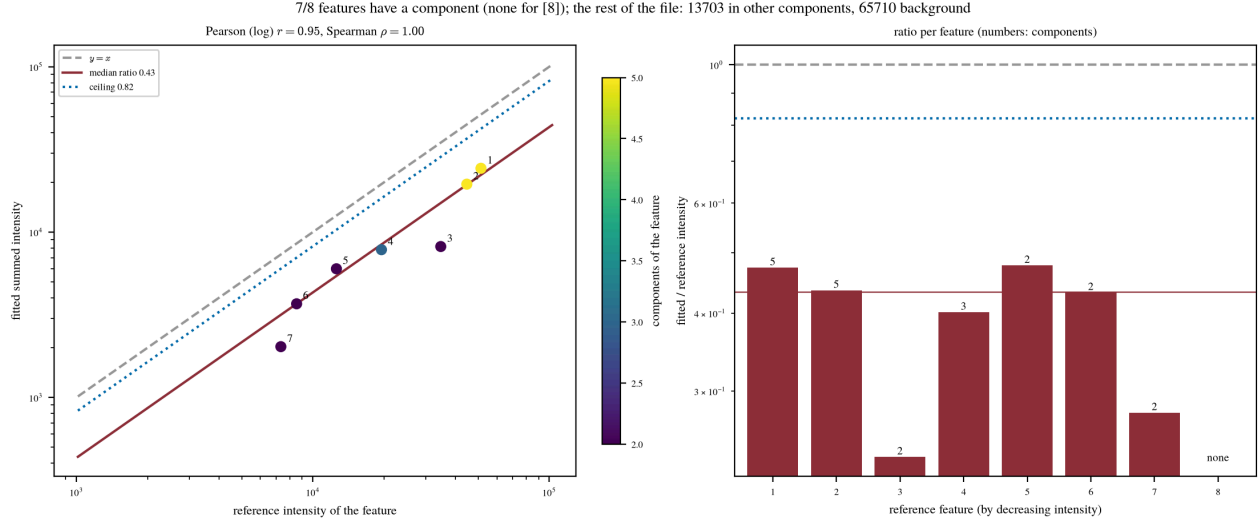


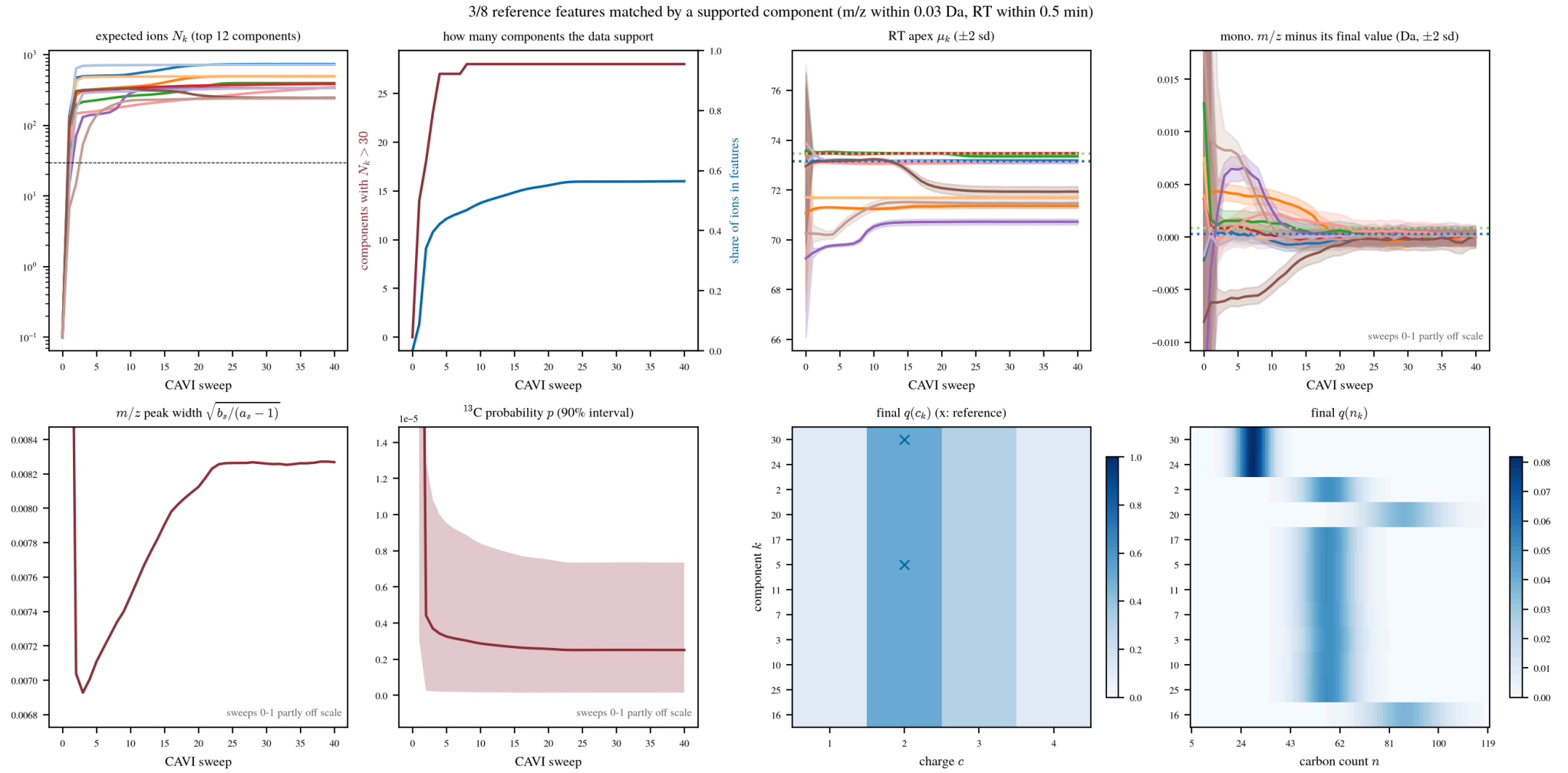
Figure 2: intensity.png: fitted against reference intensity.

Setting (OpenMS example map)	reference features matched	supported components
$K = 20 / 30 / 40$, six seeds	4.5 / 5.5 / 6.0 of 8	18–20 / 26–29 / 35–38
$K = 100 / 200$	8 / 8	– / 131
intensity, $K = 30$	Pearson (log) 0.84–0.97	fitted 0.32–0.49 times the reference (ceiling 0.82)
uniform instead of pixel start	0–1 of 8	broad width stays at 0.3 Da

Table 8: Results. Matched: a supported component at the monoisotopic mass and RT of a reference feature.

Limitations.

- **The comb is not used:** $p \approx 2 \cdot 10^{-6}$ and $\max_c q(c_k) = 0.50$ (the prior) for every seed and every K ; each isotope tooth becomes a component.
- **Few components cannot prune, many fragment:** $K = 200$ has 83 of its 131 supported components on isotope teeth; $\alpha_k < 1$ does not help (it only acts on components with about one ion).
- **Weights are not counts on real data:** the posterior is probably overconfident.

Figure 3: `metrics.png`: posterior summaries over the sweeps (sweep 0 is the random start).